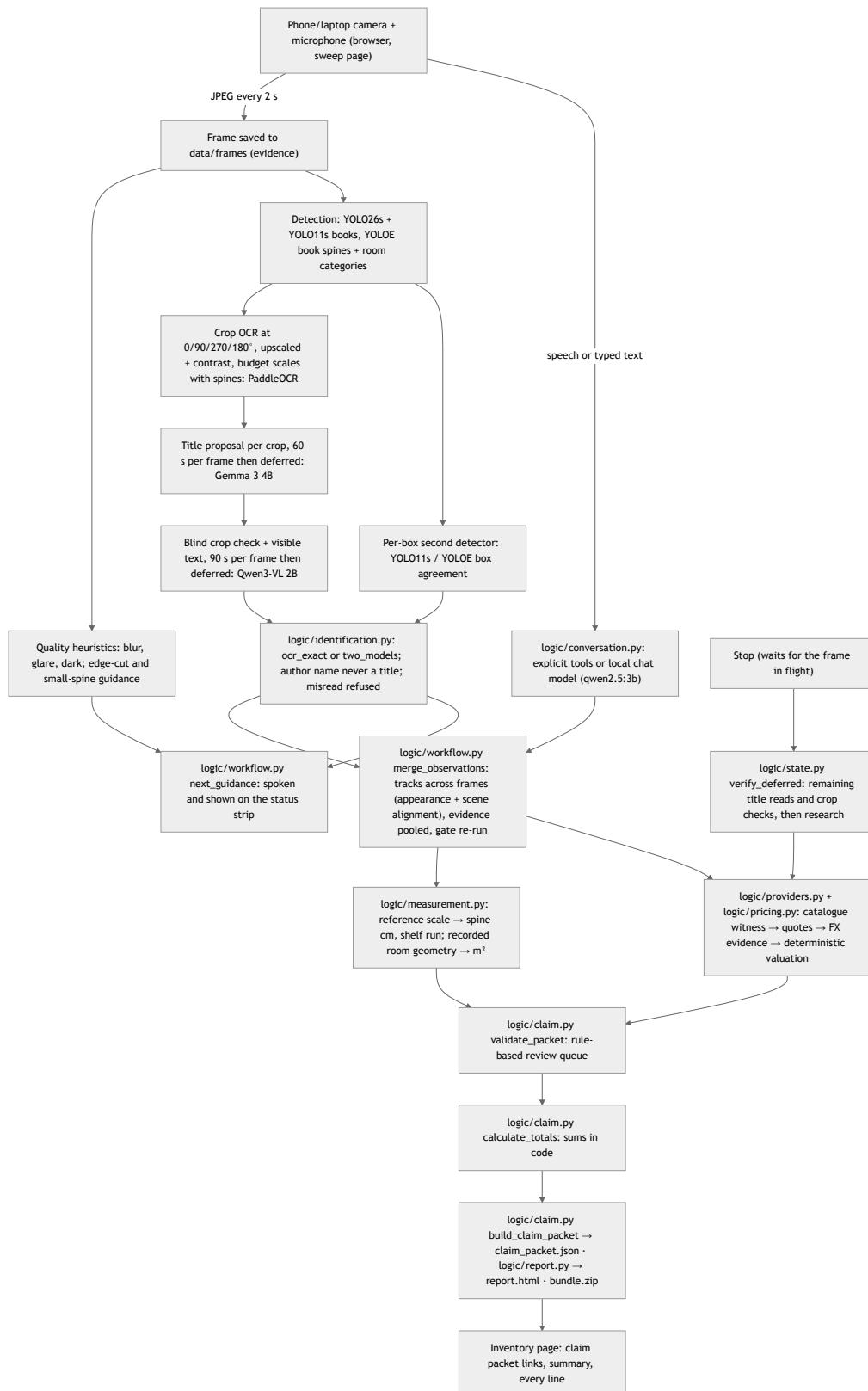


Architecture note

One continuous sweep, separate testable stages, no invented numbers.



Which model does what

Stage	Component	Role	Can it write a fact?
Capture quality	<code>logic/perception.py</code>	Pillow heuristics for blur, glare, darkness, resolution → spoken guidance	no (review notes)
Book / object boxes	YOLO26s + YOLO11s (<code>perception.detect_objects</code>) and YOLOE "book" boxes	localizes books and COCO objects; every detector that drew a box is recorded in <code>seen_by</code>	boxes only
Room categories	YOLOE26s with fused prompts (classes in <code>config.py</code>)	shelving, lamp, framed painting, rug, coffee machine, ...	category <i>proposal</i>
Second count	YOLO11s (<code>validate_boxes</code>)	per book: another detector drew the same box ($\text{IoU} \geq 0.5$) → <code>count_verified</code> ; frame-wide count equality is only a review note	agreement flag
Spine text	PaddleOCR / EasyOCR (<code>logic/ocr.py</code>)	text lines + confidence per crop, best of three rotations	OCR evidence
Title proposal	Gemma 3 4B via Ollama (<code>read_visual_title</code>)	reads the visible title; author/publisher constrained to OCR lines	proposal only
Crop check	Qwen3-VL 2B via Ollama (<code>logic/crop_verifier.py</code>)	blind category check of the crop without the detector label	agree / conflict
Identity	<code>logic/identification.py</code> (pure rules)	accepts a title only when OCR text (≥ 0.9), reader title, blind crop check and the per-book second-detector witness agree	yes (status identified)
Catalogue	Open Library (<code>logic/identification.py</code>)	resolves a work; edition only from a checksum-valid visible ISBN	work / edition metadata
Conversation	qwen2.5:3b via Ollama (<code>logic/conversation.py</code>)	answers questions with the live state; explicit commands bypass it	no

Time budgets per live frame

Stage	Budget	What happens to the rest
Detection (three detectors)	no budget; 1–25 s measured	—
Crop OCR	<code>max(10 s, 1.5 s × spines)</code> , capped at 60 s	spines beyond the budget get a note; retake closer
Title reader (Gemma)	<code>BOOK_READER_FRAME_BUDGET_S</code> = 60 s, best-evidenced crops first	marked <code>deferred</code> , read after capture stops

Stage	Budget	What happens to the rest
Blind crop check (Qwen3-VL)	<code>CROP_VERIFICATION_FRAME_BUDGET_S</code> = 90 s, wall frames first	marked <code>deferred</code> , checked after capture stops

The after-capture pass (`state.verify_deferred`) finishes the deferred reads and checks, runs the identity gate on each, and re-finishes the packet; the page shows its progress. Measured on a worn 53-spine shelf: the live frame fell from 799 s to 274 s.

Identity gate

Two evidence-bound paths, both needing a whole spine, a second detector on the same box and the blind check's agreement: `ocr_exact` (the proposal is literally in the OCR text at ≥ 0.9) and `two_models` (the title reader and the blind check, which also transcribes the spine, read the same title and the OCR fragments anchor at least a third of it in order). An author's name is never a title (reader prompt, gate rule, Open Library author search at finish). A title word the blind check read differently is refused as a probable misread, and the Open Library catalogue confirms or flags every accepted title at finish. Evidence is pooled across the frames a spine appears in, with book tracks aligned by the scene transform between views.

Frontend

Two pages in one React app (`frontend/src/App.tsx`): the sweep page (controls, camera with a status strip, live counts, chat with the agent) and the inventory page (`#inventory` : claim packet links and summary, then every line), enabled once the packet is finished. The layout is responsive from 360 px phones to wide desktops; the dev server is HTTPS so phones on the LAN get the camera and microphone.

Metric scale

- **Spines:** the claimant states a known shelf width during the sweep ("the shelf is 90 centimetres wide"); the detected shelf/book-row span in that frame gives cm per pixel, and the front-on book boxes become spine height $\times$ thickness. An explicit reference line can also be posted to `/calibration` . Method and calibration are stored on each book.
- **Room:** recorded dimensions (tape, plan or LiDAR export) against a sweep frame; rectangle or floor-plan polygon (shoelace area, perimeter $\times$ height for gross walls, shelving rectangles for shelved wall area). `room.scale_method` , `source` , `frame_ref` , `confidence` and assumptions are stored. No separate measuring pass.

Where prices come from

`logic/providers.py` holds the three retrieval paths: eBay Browse (fixed-price listings; new $\rightarrow$ replacement, used $\rightarrow$ used value; median of matching listings; comparables kept), Google Books (country list prices; ebooks flagged) and FX (dated ECB/ER-API rates for `converted: true`).

`logic/pricing.py` applies the rules without network or models and records every candidate and rejection in `price_details`. Operator-checked offers entered in the review panel carry the same evidence fields. Appraisal rules: signed/rare/antiquarian/ first editions, original art (unless the claimant says "print"), and anything at or above the configurable threshold.

Validation, review queue and totals

`logic/claim.py` rebuilds workflow findings on every change: unreadable spine, confidence below threshold, no accepted price (with the provider rejection reasons), missing source URL/date, currency mismatch, converted price without FX evidence, implausible dimensions, duplicate IDs, missing evidence frame, unknown room scale, unconfirmed coverage. Perception notes (count disagreement, OCR unavailable) and claimant statements are kept. `logic/claim.py` (totals) sums only known values; `excluded_from_totals` counts the blanks.

Latency and cost

Each stage records elapsed seconds (`pipeline_stages` per frame, `stage_runs` per workflow pass); `GET /api/sweeps/{id}/performance` summarises calls, mean and max per stage, and `time_to_packet_s` is measured from stop-capture to finish. Model API cost is zero (local inference); provider calls (eBay, Google Books, FX) are counted in `cost.provider_calls` and are free-tier.